

AKSHAY KHULE

Hands-on Principal Architect | Java/Python Full Stack · Distributed Systems · Data Platforms · Generative AI

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[Publications](#) | [GitHub](#) | [Cloud Certifications](#) | [Distributed Systems Certifications](#)



PROFESSIONAL SUMMARY

Making complex distributed systems simple.

- **Technical Leadership:** Global technical leadership across Software Engineering, Product Development, and Systems Architecture - architecting and shipping production-grade distributed systems, cloud-native microservices, petabyte-scale data platforms, and GenAI solutions across Banking, Pharma, AIOps, Manufacturing, and Cybersecurity domains, owning end-to-end High-Level and Low-Level Design for scalable, resilient systems.
- **Application Engineering:** Hands-on design and development of RESTful APIs, microservices, and serverless backends in Python (FastAPI, Pydantic, SQLAlchemy) and Java (Spring Boot, Spring Cloud, Spring Security), OAuth2/JWT with KeyCloak, RDBMS schema design, Redis, and message queues (Kafka, SQS) - engineered with service discovery, centralized configuration, resiliency patterns, distributed tracing; containerized with Docker and deployed on Kubernetes.
- **Data Engineering:** Expert in end-to-end Big Data, Data Lake, and Data Platform architecture - streaming/batch ingestion, extraction, cleaning, transformation, enrichment, governance using PySpark (Structured Streaming, Pandas UDFs), Databricks (Delta Lake, Unity Catalog), Kafka, Azure Synapse, Azure Data Factory, Synapse Datawarehouse and Microsoft Fabric; automated deployment and data-quality validation with PySpark/SQL, delivering petabyte-scale, GDPR-compliant pipelines with zero-data-loss guarantees.
- **Cloud & DevOps:** 4x AWS Certified (including Solutions Architect Professional) and Microsoft Azure Certified; hands-on with AWS and Azure cloud-native services for scalable infrastructure and applications. CI/CD and platform automation with Git, Bitbucket, GitHub Actions, Jenkins, Docker, Kubernetes (AKS/EKS), Helm, and Databricks Asset Bundles for declarative, gated pipeline deployments across dev, staging, and production environments.
- **Generative AI:** Architected the multi-agent foundation for [Unisys Intelligence Accelerator](#) (UIA), an AI-enabled delivery platform that automates workflows, orchestrates intelligent operations, and applies enterprise context across cloud, applications, data, and security environments. Built on AI orchestration frameworks such as Agno; designed and implemented production-grade RAG systems including embedding models, semantic search, hybrid search, reranking, and vector databases; integrated and deployed applications using LLMs such as OpenAI GPT, Claude, on platforms including Amazon Bedrock AgentCore and Azure OpenAI.

PUBLICATIONS & THOUGHT LEADERSHIP

- **White paper:** GenAI-Augmented Polyglot Persistence for Microservices - Toward Self-Adaptive & Explainable Data Fabrics. <https://lnkd.in/dVeegA4H>
- **Newsletter:** The Engineering Log by Akshay - ongoing LinkedIn publication on distributed systems, data engineering, and GenAI. [Read on LinkedIn](#)

CERTIFICATIONS

AWS	Solutions Architect Professional Data Analytics Specialty Security Specialty
Databricks	Databricks Certified Professional Spark Certified Associate Developer
Confluent (Kafka)	Confluent Certified Developer for Apache Kafka
Microsoft Azure	Microsoft Certified: Azure Fundamentals
DAMA	Certified Data Management Professional (CDMP)
IIT Roorkee	Generative AI Engineering Program (completed)
Anthropic	Claude 101 - Certificate of Completion

TECHNICAL SKILLS

Languages	Python, Java, Scala, SQL, PL/SQL, JavaScript, HTML
Backend & APIs	FastAPI, Pydantic, asyncio/httpx, SQLAlchemy, pytest, Flask, Spring Boot, Spring Cloud, Spring Security, Hibernate ORM, REST/OpenAPI, OAuth2, JWT, KeyCloak, EJB, JMS, React, Jenkins, Maven, Git, SonarQube
Databases & Search	Oracle, PostgreSQL, Synapse data warehouse, SQL Server, Elasticsearch, Redis, Databricks Delta Lake, Vector Search (Azure AI Search)
Big Data & Analytics	Kafka, PySpark, Databricks, Delta Lake, Unity Catalog, Hive, HDFS, Hadoop, Medallion Lakehouse, Azure Synapse Analytics, Azure Data Factory, Microsoft Fabric (OneLake), AWS Glue, Data Governance, Microsoft Purview, S3, Azure Storage, AWS Glue
Cloud & Containers	AWS, Azure, Kubernetes (AKS/EKS), Docker, Docker Compose, Helm, Databricks Asset Bundles, GitHub Actions, Azure Logic Apps, Azure Functions, AWS Lambda, API Gateway, Microsoft Sentinel (SIEM/SOAR), Log Analytics/KQL, Grafana
Generative AI	LLMs (GPT-4o, Claude), Agentic AI, Multi-Agent Orchestration (Agno), RAG, Embeddings & Vector Search, Spring AI, AI Guardrails, AWS Bedrock (AgentCore), Microsoft Foundry, Azure AI Search, Hugging Face Transformers, LLM Fine-tuning, Claude Code, OpenAI/Anthropic Python SDKs

PROFESSIONAL EXPERIENCE

Principal Architect - Unisys Corporation

May 2021 – Present

Architect (2021 – 2024); Principal Architect (2024 – Present)

Unisys CloudForte - Enterprise Monitoring & Security Product (Unisys Stealth Family)

Observability product in the CSPM/cybersecurity portfolio - monitors enterprise infrastructure and applications in real time, providing the telemetry foundation for anomaly detection, security monitoring, and threat visibility.

URL: [marketplace.microsoft.com - CloudForte for Azure](https://marketplace.microsoft.com/en-us/products/microsoft/cloudforte)

- Designed the overall product architecture, cloud platform setup, and technology choices for CloudForte. Built the event-processing layer as microservices on Azure Kubernetes Service (AKS) - consuming ~5M events/day safely (no duplicates, automatic retries, dead-letter queues for failed messages) and triggering alerts or automated fixes based on risk. Designed multi-tenant isolation across the platform using per-tenant data partitioning in storage layer.
- Integrated CloudForte with Microsoft Sentinel - streaming normalized events from the Kafka topic into Log Analytics so Sentinel's built-in detection rules apply - letting SOC teams correlate CloudForte events with threat intelligence and identity data the platform cannot see.
- Designed an event-driven GenAI incident-triage agent - Python microservices consume Sentinel incidents and invoke an Azure AI Foundry agent grounded on runbook and incident-history knowledge sources - classifying incidents, recommending fixes, and estimating impact, with remediation executing only through approved, tested runbooks and results written back to the incident. Achieved 30% MTTR reduction.

Unisys Intelligence Accelerator (UIA) - Agentic AI Platform

Enterprise multi-agent platform automating IT operations, application development, and data tasks for large enterprises.

- Designed and built the agent runtime as FastAPI microservices exposing agent capabilities via REST; containerized with Docker and deployed on Kubernetes with GitHub Actions CI/CD.
- Built a reusable Python agent harness on Amazon Bedrock AgentCore - agent lifecycle, tool/MCP registration, guardrails, and observability - with parity deployment on Microsoft Foundry; agents grounded via managed Knowledge Bases (RAG) using embeddings and vector search over enterprise documentation.
- Agentic Workflows: Designed grounded agentic workflows combining retrieval, reasoning, tool calling, structured outputs, and controlled execution; business actions remain constrained by approved runbooks, guardrails, and enterprise security boundaries.
- Multi-LLM Gateway: Built a Multi-LLM Gateway using Agno for model routing across GPT-4o, Claude, and Azure OpenAI deployments, with cost-aware and latency-aware routing, circuit-breaker based provider failover, prompt and semantic

caching, and unified observability via OpenTelemetry traces and Prometheus and Grafana dashboards for token usage, latency, and cost per model.

Banking Data Platform Modernization - Financial Services Client

- Designed the migration strategy after Microsoft deprecated the Export to Data Lake service - built a data-ingestion solution pulling Dynamics 365 F&O data from Microsoft Fabric (Fabric Link/OneLake) into an Azure Synapse dedicated SQL pool, with zero disruption to downstream financial reporting across ~100+ pipelines.
- Developed a PySpark-based solution in Azure Databricks (Delta Lake) to process Dynamics 365 F&O financial data, migrating ledger and financial-transaction processing into scalable, version-controlled Spark workloads - cutting pipeline runtime from ~3 hrs to 20 mins. Delivered through Databricks Asset Bundles (DABs) with GitHub Actions CI/CD. Implemented Databricks Lakehouse Federation to query external SQL sources in place through Unity Catalog, avoiding redundant ingestion pipelines.
- Forward Deployed Engineering: Operated in a forward deployed engineering capacity, embedded directly with the client's finance and platform teams to translate business requirements into production-ready pipelines on the client's live environment, iterating rapidly with stakeholders and owning delivery from design through production cutover.

Architect - Cognizant (Client: Novartis)

July 2018 – May 2021

Adverse Event resolution product for Novartis - analyses adverse events and predicts resolutions.

URL: <https://ae-service.novartis.net/pae-coreui>

- Architected and delivered a full-stack, event-driven system - extracting adverse-event data from Salesforce Social Studio streams and partner APIs, then cleaning, normalizing, deduplicating, and enriching it through microservices, AWS Lambda, API Gateway, and SQS over REST - driving cloud adoption that saved nearly \$1M annually.
- Led the Strategic Data Platform build-out, moving the stack from synchronous to asynchronous processing (SQS + Lambda) and designing an S3 data lake consolidating RDBMS and file-system sources with schema normalization and entity matching - solving critical data-latency problems for the data-ingestion product while driving Data Security, HIPAA, and GDPR compliance. Contributed hands-on across the full stack - UI, APIs, backend, and data pipelines.

Technical Manager - Capgemini (Client: Morgan Stanley)

Feb 2017 – June 2018

- Owned end-to-end product development - architecture, design, development, release, and maintenance of Spring Boot microservices: defined service boundaries and API contracts, implemented REST controllers, service and DAO layers, and designed entities and the underlying data model, and full-stack development using React JS.
- Designed the microservices architecture with clear separation of concerns - stateless services, layered design (controller/service/repository), centralized configuration, and inter-service communication over REST - enabling independent development, testing, and release of services.

Technical Lead - Persistent Systems (Clients: Thermo Fisher, Proofpoint)

Sept 2014 – Jan 2017

Thermo Fisher - Ion Reporter: NGS product automating sequencing, analysis, and reporting. URL:

<https://ionreporter.thermofisher.com>

- Led backend development of REST APIs and Spring/Hibernate persistence layers for the sequencing analysis workflow; designed batch processing for large result sets, cutting report-generation time.

Proofpoint - Enterprise email security and compliance platform (threat protection, DLP, archiving).

- Built high-volume message-processing and policy-management services (Spring, Hibernate, REST) for threat-detection and DLP/archiving modules, with idempotent processing, RBAC, and audit trails for regulated customers.

Earlier Roles - Worldline (BNP Paribas), Polaris (JPMorgan), HAL

Jan 2009 – Aug 2014

- Built Spring/Hibernate transaction engines for SEPA direct-debit and credit-transfer processing (Worldline); designed EJB/JMS multi-tier enterprise platforms for JPMorgan's TLIP liquidity application (Polaris); developed an in-house inventory and supply-chain ERP with batch processing (HAL).

KEY ENGINEERING HIGHLIGHTS

Microservices & Backend

- Backend Stack Ownership: Took complete ownership of the backend stack - from development through cloud deployment - for 30+ microservices serving enterprise applications across multiple organizations; ensured resiliency via circuit breakers and zero-downtime deployments on Kubernetes (AKS/EKS) with auto-scaling, rolling updates, and self-healing.
- Event-Driven Consistency: Built SQS-based microservices using the transactional outbox pattern - saving business data and the outgoing event in one database transaction before publishing.
- Python Async Services: designed FastAPI services with asyncio/httpx for non-blocking I/O, Pydantic request/response validation, OpenAPI-driven contracts, and Kafka/SQS consumers in Python for asynchronous, distributed processing.

Distributed Systems Design

- High-Throughput Partitioning: diagnosed and fixed hot-partition skew in high-velocity Kafka pipelines by redesigning topic partition keys (entity-level composite keys).
- Fault-Tolerant Design: designed cloud data-ingestion services for partition tolerance - idempotent, retry-safe operations with exponential backoff and jitter, plus dead-letter queues.

Massive-Scale Data Ingestion, Engineering and Governance

- Petabyte-Scale Ingestion and Streaming: architected a high-velocity Spark ingestion engine handling petabytes of records and event-driven Kafka pipelines with fault-tolerant consumer-group services (Java and Python), cutting data availability to minutes. Implemented entity resolution and golden-record creation (MDM) with Azure Purview for governance and lineage.

DevOps & CI/CD

- Infrastructure as Code: architected Databricks Asset Bundles to declaratively define, version, and promote data workloads across environments - integrated with secure GitHub Actions pipelines on VNet-isolated self-hosted runners with strict network boundaries, environment-specific variable injection. Used ADF arm template for Pipeline deployment.

Generative AI

- Applied AI for Data: Applied LLMs, embeddings, and vector search to data engineering problems - incident classification, entity matching, and automated data enrichment.

AWARDS & RECOGNITION

- **Unisys All-Hands Recognition:** recognized four times at company-wide All-Hands meetings for high-impact work across distributed systems and data engineering initiatives.
- **“You Rock” Award Nomination - Persistent:** nominated by peers and leadership for consistent technical excellence.

EDUCATION

- Bachelor of Engineering (B.E.), Computer Science - V.Y.W.S. College of Engineering, Badnera, Amravati University (1st Division).
- Generative AI Engineering Program - IIT Roorkee.